



Course Code: CS-11303
Credit Hours: 4 (3,1)
Batch: Fall-2025

Course Title: **Operating System**
Lecture Hours/Week: 3
Pre-Requisites: **None**

Complete Analysis of Effectiveness of Course and Level of Skills Ensured in Operating System

The Operating Systems course provides foundational understanding of system-level computing principles.

Progressive Learning Structure

The course follows a layered approach:

1. OS Introduction & Structure
2. Process & Thread Management
3. CPU Scheduling
4. Synchronization & Deadlocks
5. Memory Management
6. File Systems
7. Storage & I/O Systems

Technological Alignment

Students gain theoretical grounding aligned with:

- Linux-based system architecture
- Virtualization platforms
- Cloud infrastructure environments
- Modern distributed systems

Emerging Paradigms Covered

Emerging Paradigm	Course Integration
Virtualization	Hypervisor & VM concepts
Cloud OS	Resource abstraction
Containerization	Process isolation
Cybersecurity	Access control & protection
Multi-core Processing	Thread scheduling

Industry Relevance

Students become capable of:

- Understanding Linux internals
- Designing efficient system-level solutions
- Preparing for DevOps, SysAdmin, SOC Analyst and other Cybersecurity roles

The course ensures strong analytical, algorithmic, and system-design skills aligned with PLOs.